The Boussinesq–compressible gap in two-dimensional steady Rayleigh–Bénard convection is linear to leading order in the relative temperature contrast: a factorised physics prior for surrogates

Author(s)[a]

[a]*Affiliation(s)*

Abstract

Boussinesq-trained surrogates do not transfer across contrast. For steady two-dimensional Rayleigh–Bénard convection one parameter governs the difference, the contrast $\varepsilon = \Delta T/T_0$. The Froude number is not free ($Fr^2 = \varepsilon$), and a 24-fold change in Ma moves the $O(Ma^2)$ correction by less than 10^{-5} in the retained low-Mach equations. Over 627 paired solutions ($\varepsilon = 0.02$–1.9), the relative correction obeys $R = A\varepsilon^m$ with $m = 1.023 \pm 0.018$, 0.995 ± 0.019, 1.024 ± 0.077 and 1.009 ± 0.015 for (u_x, u_y, θ, p), a vanishing intercept and ε^2 curvature; the θ amplitude is 0.010 wall-normalised, 0.034 zero-mean. It reproduces on another group's public data ($\varepsilon \equiv 1.2$), survives a second geometry and $Pr = 4.38$, and stays within 0.11 of unity in a twelve-sample 3-D check ($Ra = 10^4$, $res = 16$, $\varepsilon \leq 1.2$). At $\varepsilon = 1.2$ the integral Nusselt number changes by 0.6%, the local wall flux by 4.25%. The prefactor is not universal: in θ it collapses on the peak Boussinesq velocity across generators. Because $D = \varepsilon P(X)$, a surrogate predicts P and multiplies: with ε as input it improves 1.07–1.15× over a FiLM-conditioned network; without it, it still beats an ε-blind network by 1.89× and reuses single-ε public data within a boundary family. One inference takes 6.3 ms; replacing the compressible member cuts a paired solve from 33.9 s to 11.7 s (2.9×) and a 100-contrast sweep by 280.

Keywords: natural convection, Boussinesq approximation, low-Mach compressible flow, physics-informed surrogate, neural operator, break symmetry

Nomenclature

Roman		*Greek*	
A	prefactor of $R = A\varepsilon^m$	β	thermal expansion coefficient
D	correction $Y - X$, the surrogate target	γ	ratio of specific heats
e	internal energy	ΔT	wall-to-wall temperature contrast
Fr	Froude number, $Fr^2 = \Delta T/T_0$	ε	relative contrast, $\Delta T/T_0$
L	side length of the cavity	θ	normalised temperature
m	scaling exponent	κ	curvature, $R = A\varepsilon(1 + \kappa\varepsilon)$
Ma	Mach number	ρ	density
n	number of samples in a data set or split	$\boldsymbol{\sigma}$	viscous stress tensor
Nu	Nusselt number		
p	pressure		
P, Q	first- and second-order correction fields		
Pr	Prandtl number		
R	relative correction, $\|Y - X\|/\|Y\|$		
Ra	Rayleigh number		
Re	Reynolds number, $(Ra/Pr)^{1/2}$		
T	temperature, T_0 its reference value		
$\boldsymbol{u}$	velocity, (u_x, u_y)		
$\hat{u}$	peak Boussinesq speed		
X, Y	Boussinesq and compressible solutions	$\mathcal{L}$	linearised Boussinesq operator

1. Introduction

Steady natural convection is a workhorse of thermal engineering and it is expensive. A converged pair of two-dimensional solves on a 64^2 mesh costs 34 s of wall clock on an eight-way desktop (Sec. 5.12); the three-dimensional problems that engineering actually needs cost hours to days. That is the standard motivation for surrogate modelling, and the standard response is to train a network that maps a cheap model's output to an expensive model's output [1, 2].

For convection the two models are not two arbitrary fidelities: they are the Boussinesq (Oberbeck–Boussinesq) equations and the compressible low-Mach equations. Boussinesq is exact only in the limit of vanishing relative temperature contrast; the corrections are $O(\beta\Delta T)$ and have been characterised asymptotically for decades [3, 4]. Recent work makes the pair directly learnable. Menglik et al. [5] publish paired Boussinesq→compressible steady solutions in two dimensions (4000 training pairs plus held-out splits, CC BY 4.0) and train a Fourier neural operator [6] to predict the compressible field from the Boussinesq field. That data set—which we use as an independent validation set in Secs. 5.7–5.9—contains exactly one temperature contrast: $\Delta T = 1.2$ at $T_0 = 1$.

The single-contrast design is the practical crux. A surrogate fitted on such a data set has learned the correction at one operating point. At any other temperature contrast the Boussinesq input already differs from the input seen in training, so the correction the network memorised is the wrong correction—not merely an inaccurate one—and the only remedy on offer is to generate a second paired data set, i.e. to pay again exactly the cost the surrogate was meant to remove. We show in Sec. 5.3 that this is not a modelling failure that more capacity or a better architecture can repair: it is an information barrier, and we measure it.

This paper makes the barrier explicit and then removes it.

What governs the gap. For the non-dimensionalisation used here the Froude number is not free: $Fr^2 = \beta\Delta T = \Delta T/T_0$ identically, and the Mach number enters the equations only at $O(Ma^2)$ within the retained low-Mach equations (Sec. 2.2, and measured: a factor-24 change in Ma moves the correction by less than 10^{-5}). The break-symmetry group is therefore one-dimensional, and it has a name: $\varepsilon = \Delta T/T_0$.

What the gap looks like. Expanding the compressible member in ε gives an $O(1)$ system that is term-by-term the Boussinesq system, plus an $O(\varepsilon)$ source produced by the density-modulated inertia and by the non-solenoidal velocity field $\nabla \cdot \boldsymbol{u}_1 = \nabla\cdot(\theta_0\boldsymbol{u}_0)$. Because the linearised operator is ε-independent, the correction factorises, $Y - X = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots$, with a shape P that does not rotate as ε moves (Sec. 3). Measured over two decades of ε (627 samples, Sec. 5.1) the relative correction obeys $R = A\varepsilon^m$ with $m = 1.02$ in all four channels (a window-averaged exponent: the same 627 points give 1.23–1.32 when fitted from $\varepsilon = 0.8$ upwards, the ε^2 curvature of Sec. 5.1), a vanishing intercept, and a prefactor whose weak Rayleigh-number trend reproduces in a second geometry and in a third Prandtl number.

What it buys. Because the ε dependence can be written outside the network, a surrogate can predict $P(X)$ and multiply by ε afterwards. Supplying ε as an input as well gives the largest gain—1.07× in extrapolation for the linear form and 1.15× for the quadratic form, relative to the strongest conditional baseline (Sec. 5.5)—while removing it from the input costs a little accuracy but is what lets a network trained on a public data set that exists at a single ε be applied, without retraining, to our $\varepsilon \in [0.2, 1.9]$, where it reaches relL2 = 0.093 (u_x) against 0.144 for the plain model at the same budget (Sec. 5.9). The prior is also the cheaper model when data is scarce (Sec. 5.6: 1.4× better than an ε-augmented baseline at $n = 50$, crossover near $n \approx 300$), the network discovers the ε^2 curvature of the expansion on its own (learned $\kappa = 0.22$–0.42 in all four channels), and the same prior transfers unchanged to a Fourier neural operator.

1.1. Related work

Reduced-order models replace the discretised equations by a small number of modes; parametric dynamic mode decomposition is a representative recent example [7]. Graph convolutional networks exploit the mesh and have been applied to natural convection in annuli [8] and to geometry-adaptive steady convection with physics-informed losses [9]. Operator learning removes the mesh-resolution constraint: DeepONet approximates the solution operator from branch and trunk networks [10], physics-informed DeepONets add the residual to the loss [11], and Fourier neural operators (FNOs) learn a global convolution in Fourier space [6]. FNOs have since been used for sparse-field reconstruction [12], real-time heat-pipe startup prediction [13], and operator-level uncertainty quantification in cooling-channel optimisation

[14]. Physics-informed neural networks [15] and their heat-transfer variants [16, 17] take the opposite route: they solve one instance by minimising a PDE residual rather than learning a family of instances. Reviews of the numerical analysis and of the application landscape are given by De Ryck and Mishra [18], Meng et al. [2] and Luo et al. [19].

A physics prior can enter as a residual penalty, as a conservation law built into the architecture, or as an algebraic factorisation of the output. The first two are well developed; the third is the subject of this paper. Residual-based corrector operators solve a linear variational problem for the residual of a trained operator and add the solution to its prediction; the correction is deterministic, costs one linear solve per query, and improves accuracy without retraining the base operator [20, 21]. Physics-informed extrapolation of deep operators has been demonstrated by adding sparse observations or residual terms [22], and transfer operator learning addresses distribution shift between parameter regimes [23]. Multi-fidelity surrogates combine cheap and expensive data through LSTM corrections [24], feature-adjacent physics-informed learning [25] or transfer-learned FNOs [26]. These methods reduce the amount of high-fidelity data, but they still learn the parameter dependence from samples. For buoyant convection, the Oberbeck–Boussinesq approximation and its validity limits are classical [3, 4], and the review of Ahlers et al. [27] summarises the heat-transfer and large-scale dynamics of turbulent Rayleigh–Bénard convection. Non-Oberbeck–Boussinesq (NOB) effects have been measured under strong temperature contrast [28], in cryogenic helium [29] and in a water-filled differentially heated cavity [30]. At $Pr = 0.71$ and large contrast they are not a small correction: they grow cold-corner vortices and can reverse the large-scale circulation [31]. On the numerical side, low-Mach compressible formulations for enclosed thermal convection have been developed and compared with the Boussinesq limit [32, 33], and the rigorous derivation of the Oberbeck–Boussinesq approximation has recently been revisited [34]. What is missing is a quantitative, closed-form account of how the *difference between the two solutions* scales with the temperature contrast, and a surrogate design that uses that scaling. These studies fix ε and compare the two solutions; none extracts the ε dependence as a reusable prior. The present work differs in that the parameter dependence is derived, and only the spatial shape of the correction is learned.

Table 1 places the resulting prior among the four families.

Contributions.

1. We identify the exact, one-dimensional break-symmetry parameter of the Boussinesq–compressible pair and show the group is one-dimensional rather than two (Sec. 2.2).
2. We derive the correction's factorisation from an ε-expansion and verify all five of its predictions numerically, including the divergence relation (Sec. 3).

3. We calibrate the resulting law on 627 paired samples spanning two decades of ε, in two geometries and two Prandtl numbers, and validate it on an independently generated public data set—including the secondary Rayleigh-number trends (Secs. 5.1, 5.7, 5.10, 5.13).
4. We measure what the gap does to the wall heat transfer: at $\varepsilon = 1.2$ the integral Nusselt number moves by 0.6%, while the local wall flux is redistributed by 4.25% (Sec. 5.8).
5. We turn the law into a surrogate design rule ($D = \varepsilon P(X)$), measure what it is worth (1.07–1.15× over the strongest conditional baseline), and show it makes an existing single-ε public data set reusable, with the end-to-end saving measured per pair (2.9×) and for a 100-contrast sweep (2.8×10^2) (Secs. 5.4–5.9, 5.12).

All results are two-dimensional except the twelve-sample three-dimensional check of Sec. 5.14. Section 6 states the measured cost of the three-dimensional problem on the same hardware and gives the concrete reasons we do not claim a 3-D sweep.

Table 1: Where the present prior sits among physics-informed operator methods. "Derived" is obtained in closed form from the equations; "learned" is supplied by the network. Rows: post-hoc correctors [20, 21], physics fine-tuning and multifidelity [22], transfer operator learning [23], dimension-informed operator [35].

family	derived	learned	target data	per query	law
post-hoc correctors	linear residual	none	no	one linear solve	no
physics fine-tuning	residual / data	correction	yes	no	no
transfer operator	none	operator	yes	no	no
dimension-informed	equivariance	operator	no	no	no
this work	ε expansion	shape P	no	no	m, κ

2. Problem, discretisation and the break-symmetry parameter

2.1. Two members per sample

Every sample is a *paired* steady solution of one physical configuration with two equation sets on the same mesh ($res = 64$, $L = 1$, $Pr = 0.71$, $\gamma = 1.4$). The high-fidelity member Y is the compressible low-Mach formulation in $\boldsymbol{w} = (\boldsymbol{u}, T, \rho, p)$:

$$\nabla \cdot (\rho \boldsymbol{u}) = 0, \tag{1}$$

$$\nabla \cdot (\rho\, \boldsymbol{u} \otimes \boldsymbol{u}) = -\nabla p + \frac{1}{Re} \nabla \cdot \boldsymbol{\sigma} - \frac{1}{Fr^2} \rho \boldsymbol{g}, \tag{2}$$

$$\nabla \cdot (\rho E \boldsymbol{u}) = \frac{\gamma}{Re\, Pr} \nabla^2 e + \nabla \cdot (\boldsymbol{P}_{\text{work}} \boldsymbol{u}) + O(Ma^2), \tag{3}$$

$$\rho e = \gamma Ma^2 p + 1, \qquad Re = (Ra/Pr)^{1/2}. \tag{4}$$

The low-fidelity member X is the Boussinesq system in $(\boldsymbol{u}, T, p)$, written in the *same* free-fall scales:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{5}$$

$$\boldsymbol{u} \cdot \nabla \boldsymbol{u} = -\nabla p + \left(\frac{Pr}{Ra}\right)^{1/2} \nabla^2 \boldsymbol{u} + \theta \hat{\boldsymbol{g}}, \tag{6}$$

$$\boldsymbol{u} \cdot \nabla T = (Ra\, Pr)^{-1/2} \nabla^2 T, \qquad \theta = \frac{T - T_0}{\Delta T}. \tag{7}$$

Both are solved with Taylor–Hood-type mixed elements, damped Newton iterations and a MUMPS factorisation; steady states are reached by an adaptive march with a convergence guard. The temperature boundary data are random Fourier wall temperatures with per-wall mode dials.

2.2. Fr is not a parameter, Ma is negligible, ε is the axis

Table 2 lists the three candidates for a break-symmetry axis and what rules each one out.

Table 2: The three candidate break-symmetry parameters. Only ε survives as a free axis in the present non-dimensionalisation. The Ma row is a consistency check on the retained equations, which drop the $O(Ma^2)$ term.

candidate	evidence	conclusion
Fr	the generators state $Fr^2 = \beta\Delta T = \Delta T/T_0$ identically; the released 2-D code pinned $Fr = 1.095$, correct only at $\Delta T = 1.2$	not free; pinned-Fr sweeps of $T_0/\Delta T$ are an artefact
Ma	fixing Ra, T_0, ΔT and changing Ma by 24× leaves the correction amplitude unchanged to five significant figures	$O(Ma^2) \approx 10^{-6}$: unmeasurable
$\varepsilon = \Delta T/T_0$	same boundary-condition seed stream, ΔT swept	the clean, one-dimensional break-symmetry axis

2.3. The low-fidelity solution is exactly ε-blind

θ and $Pr\,Ra$ are the only combinations that appear in the Boussinesq weak form; neither T_0 nor ΔT appears alone. Hence, with the thermal data prescribed in θ and Ra fixed, X cannot depend on ε. This was verified to solver precision: across 20 configurations × 5 values of ε (same Ra, bit-identical θ boundary stream), the maximum relative deviation of X is 9.77×10^{-8}.

> **Corollary.** The correction $Y - X$ is information that is *absent from the surrogate's input*: no amount of capacity or training data can recover it from X alone. The limit is informational, not architectural (Sec. 5.3).

3. Asymptotic theory: why the correction factorises as $\varepsilon P(X)$

Write $T = T_0(1 + \varepsilon\theta)$ with $\theta = O(1)$. With $\rho = 1/T$ (the equation of state and initial condition used here) and $Fr^2 = \varepsilon$, the buoyancy source is

$$\frac{\rho \boldsymbol{g}}{\varepsilon} = \frac{\boldsymbol{g}}{\varepsilon} - \theta \boldsymbol{g} + \varepsilon\theta^2 \boldsymbol{g} + \ldots, \tag{8}$$

so the $O(1)$ balances are

$$\nabla \cdot \boldsymbol{u}_0 = 0, \tag{9}$$

$$\boldsymbol{u}_0 \cdot \nabla \boldsymbol{u}_0 = -\nabla \tilde{p}_0 + Re^{-1} \nabla^2 \boldsymbol{u}_0 + \theta_0 \hat{\boldsymbol{g}}, \tag{10}$$

$$\boldsymbol{u}_0 \cdot \nabla \theta_0 = (Re\, Pr)^{-1} \nabla^2 \theta_0, \tag{11}$$

which is term-by-term the Boussinesq system solved as X ($Re^{-1} = (Pr/Ra)^{1/2}$). The correction at $O(\varepsilon)$ is driven by the density modulation:

$$\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0), \tag{12}$$

$$\nabla \cdot (\boldsymbol{u}_1 \otimes \boldsymbol{u}_0 + \boldsymbol{u}_0 \otimes \boldsymbol{u}_1 - \theta_0 \boldsymbol{u}_0 \otimes \boldsymbol{u}_0) = -\nabla \tilde{p}_1 + Re^{-1} \nabla^2 \boldsymbol{u}_1 + \theta_1 \hat{\boldsymbol{g}} + \theta_0^2 \hat{\boldsymbol{g}}. \tag{13}$$

Since $\mathcal{L}$, the linearisation of the Boussinesq operator about X, does not contain ε,

$$D = Y - X = -\varepsilon \mathcal{L}^{-1} S_1[X] - \varepsilon^2 \mathcal{L}^{-1} S_2[X] + \cdots = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots \tag{14}$$

Five predictions follow, and all five hold; Table S1 of the supplementary material lists them and their measured values.

The drift of the divergence check with ε (0.994 → 0.901) is the truncated ε^2 term, which is exactly what the fifth row of Table S1 measures independently.

Domain of validity. $\varepsilon < 2$, since the cold wall $T_c = T_0 - \Delta T/2$ must be positive.

4. Data sets and protocols

Table 3 lists the data sets.

Surrogates use residual learning $\hat{Y} = X + \mathcal{N}(\cdot)$, a U-Net of width 32 (or an FNO2d with 12 modes, width 32, four layers, 1.19 M parameters), Adam with cosine annealing, 200 epochs, batch 16, and targets normalised per channel. Inputs and targets are scaled per channel by standard deviations of the training split; in the cross-data-set transfer experiment the target pool is scaled by the standard deviation of its own inputs, which uses no labels. Two splits are used throughout: extrapolation (train $\varepsilon \leq 1.3$, test $\varepsilon > 1.3$, $n = 218$) and interpolation (stratified 2:1 in ε, $n = 200$). Errors are the per-channel relative L_2 error against Y; "floor" means $\hat{Y} = X$.

Tables that support the argument but are not needed in its flow—the asymptotic predictions, the two-parameter fits, the exact single-ε transfer matrix, the learning curve, the public data splits, the prefactor–velocity law and the map from scripts to reported numbers—are collected in the supplementary material; they are cited where they are used.

Table 3: Data sets used in this study. The public sets are from Menglik et al. [5].

set	n	what varies	purpose
random-BC	600	$Ra \sim \log U[10^2, 10^6]$, $\varepsilon \sim U[0.2, 1.9]$, 134 wall-mode families	calibrate the law, train surrogates
$\varepsilon \to 0$ checks	27	$\varepsilon = 0.02/0.05/0.10 \times Ra = 10^3/10^4/10^5$	test $R(0) = 0$
paired	20×5	only ε; Ra and the θ boundary stream fixed within a configuration	exact ε-blind floor, shape invariance
aspect 2:1	104	same law, geometry 2×1	external validity
tight rerun	10	the 10 worst-converged samples, tolerance $10^{-6} \to 10^{-8}$	noise control
public, random walls [5]	$4000 + 2 \times 200$	$\varepsilon \equiv 1.2$; the two held-out splits differ in the range of the wall dials	external validation, cross-data-set transfer
public, classical cavity [5]	5000 + 200	constant hot/cold side walls, adiabatic top and bottom, $\varepsilon \equiv 1.2$	external validation of the prefactor
$Pr = 4.38$ (this work)	104	$Pr = 4.38$; BC/Ra/ε draws bit-identical to 104 of the random-BC samples	test of the fluid axis

Temperature normalisation. Eq. (7) defines the zero-mean variable $\theta = (T - T_0)/\Delta T$. The public release and our own pipeline store T/T_0 and evaluate every temperature error on the wall-normalised $\theta_w = (T - T_c)/\Delta T \in [0, 1]$, which differs from θ by the constant $1/2$. All R_θ, A_θ and p_θ quoted below are in the wall-normalised convention. The same field difference reads 0.0220 wall-normalised against 0.0869 zero-mean on the public random-wall test split, and 0.0836 against 0.1628 on the classical cavity; the two conventions must not be mixed. In Table 4 the θ amplitude is therefore listed in both, 0.010 wall-normalised and 0.034 zero-mean.

5. Results

5.1. The ε-law, calibrated over two decades

Fig. 1 shows the collapse and Table 4 collects the fits.

Table 4: Power-law fit $R = A\varepsilon^m$ over the full two-decade window (627 samples). The intercept is fitted separately on the small-ε subset ($\varepsilon \le 0.5$). For θ the amplitude is given in both conventions, wall-normalised/zero-mean.

channel	$R = A\varepsilon^m$	A	R^2	intercept ($\varepsilon \le 0.5$)
u_x	$m = 1.023 \pm 0.018$	0.147	0.84	$+0.0003 \pm 0.0047$
u_y	$m = 0.995 \pm 0.019$	0.193	0.82	$+0.0030 \pm 0.0062$
θ	$m = 1.024 \pm 0.077$	0.010/0.034	0.22	-0.0003 ± 0.0012
p	$m = 1.009 \pm 0.015$	0.148	0.87	$+0.0016 \pm 0.0030$

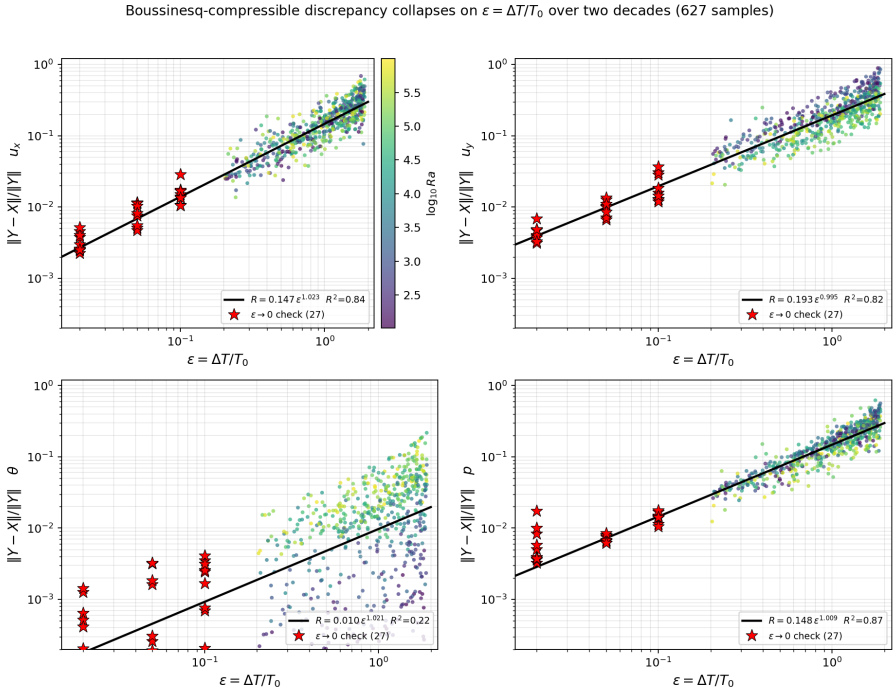


Figure 1: The Boussinesq–compressible discrepancy collapses on $\varepsilon = \Delta T/T_0$ over two decades. Points: 600 random-boundary-condition samples, coloured by $\log_{10} Ra$; stars: the 27 independent small-ε checks. Lines: the power-law fits of Table 4.

The exponent is window-averaged, and the curvature is real. The fit above uses the full two decades, including the 27 check points at $\varepsilon \leq 0.1$ where R is closest to its linear limit. Fitting only $\varepsilon \geq 0.8$ gives $m = 1.23 \pm 0.08/1.25 \pm 0.08/1.21 \pm 0.34/1.32 \pm 0.06$, and fitting each of the 20 paired configurations separately over $\varepsilon \in [0.4, 1.9]$ gives a median $m = 1.31/1.17/1.21/1.45$ (median $R^2 = 0.98$, `code/audit_v2.py`). That is the signature of $R = A\varepsilon(1 + \kappa\varepsilon)$ rather than of an exact power: a power law fitted to a curved function over a window that excludes small ε returns $m \approx 1 + \kappa\langle\varepsilon\rangle$. The quadratic fit $R = a\varepsilon + c\varepsilon^2 + b$ gives $c > 0$ in all four channels (u_x: $+0.0513 \pm 0.0133$, 3.9σ; u_y: $+0.0695$; θ: $+0.0060$; p: $+0.0539$) with $\kappa = c/a = 0.70/0.83/0.42/0.73$. We therefore state the law as first order in ε, $R = A\varepsilon(1 + O(\varepsilon))$, quote $m = 1.02$ as the number a wide-window fit returns, and note that the network recovers the ε^2 curvature with the same sign and order of magnitude (Sec. 5.5, $\kappa = 0.22$–0.42) and by the mode expansion (Sec. 3, $\varepsilon^1 + \varepsilon^2$ carry 99.0% of the correction energy).

- ε-scaling is geometry-independent; the prefactor A is not ($0.147 \to 0.175$ for u_x, $0.148 \to 0.222$ for p, $0.193 \to 0.175$ for u_y, $0.010 \to 0.008$ for θ).
- Boundary-condition families barely matter *inside this family*: 134 wall-mode families, 133 free dummies. Adding them to the ε-only fit raises the adjusted

R^2 by 2.3 pp (p), 2.9 pp (u_y), 3.3 pp (u_x) and 8.5 pp (θ)—the ε-scaling is not an artefact of which wall modes were drawn (`code/audit_numbers.py`).

- The θ channel needs one extra, cheaply available parameter, the advection strength of the base flow $\chi = \|\boldsymbol{u}\|\|\nabla\theta\|/\|\theta\|$: $R_\theta = \varepsilon(-0.0037 + 0.226\chi)$, R^2 0.185 $\rightarrow$ 0.778 (Spearman 0.902), i.e. $R_\theta \approx 0.23\varepsilon\chi$. For $u_x/u_y/p$ adding χ does nothing. A single-input form of the same statement, $R_\theta/\varepsilon = c\hat{u}^a$, is calibrated on two independent generators in Sec. 5.7 and carries the same information with $R^2 = 0.90$.

- A pressure-channel preprocessing bug had to be fixed first (Sec. 5.2).

5.2. A pressure-channel artefact, and why it mattered

Pairing the two members' *raw* pressures mixes units: the compressible raw pressure carries a hydrostatic ramp 12× larger than its dynamic part, and the two weak forms use opposite pressure sign conventions. Before the fix the per-sample median of $|X_p|/|Y_p|$ was 0.065 and that of $\cos(X_p, Y_p)$ was -0.105, and the p-channel "law" was a constant ($R_p \approx 1.0$). Applying the recipe used by the public data set—subtract the mean, remove a least-squares linear-in-y fit, flip the sign of the Boussinesq member—gives 0.960 and cos = +0.991 under the same median convention, against 0.929 and +0.986 for the public training set, i.e. we now match their pairing to within 3% (`code/audit_v2.py`). The channel then obeys the same law as the velocities ($A = 0.148$, $m = 1.009$). All p-channel numbers reported here are post-fix.

5.3. The ε-blind ceiling, measured exactly

Within a configuration X is identical for all ε, so an ε-blind model must emit one field F for every ε it is applied at (Fig. 2, Table 5). Two reference protocols are exact rather than fitted: (A) train at the one ε a published data set offers (here $\varepsilon = 1.2$) and apply it everywhere, error $\|Y(\varepsilon) - Y(1.2)\|/\|Y(\varepsilon)\|$; (B) the single field that best serves the whole range, the ε-mean of Y. (B) is a least-squares compromise over the ε-range, not a per-ε lower bound—a field tuned to $\varepsilon = 0.8$ gives 0.067 at $\varepsilon = 0.4$, better than (B)'s 0.170 there, and below $\varepsilon \approx 0.8$ doing nothing beats both. Median over 20 configurations, channels $u_x/u_y/\theta/p$:

Protocol (A) is what a single-ε public data set forces: 25% (u_x), 30% (u_y) and 35% (p) error at $\varepsilon = 1.9$, 8% (u_x) already at $\varepsilon = 0.8$, and—the point for practice—at $\varepsilon = 0.4$ "do nothing" is *better* than it, and the two cross between $\varepsilon = 0.4$ and 0.8. Protocol (B), the least-squares compromise over the range and therefore the best a single ε-blind field can do on average, still leaves 22.5/28.6/28.1% at $\varepsilon = 1.9$. The whole effect is one line of arithmetic, and the paired set measures it exactly (`code/exp_paired_final.py`; the diagonal is identically zero because the same ε is the same solution): the full 5×5 matrix is Table S3 of the supplementary material.

The error is set by the distance in ε and by nothing else—which is precisely the quantity $\varepsilon P(X)$ puts back into the model.

Table 5: The exact ε-blind floor. Protocol (A) is what a single-ε public data set forces; protocol (B) is the best single field over the whole range. Entries are medians over 20 configurations, ordered $u_x/u_y/\theta/p$.

ε	do nothing ($\hat{Y} = X$)	(A) train at $\varepsilon = 1.2$ only	(B) ε-mean field
0.4	0.060/0.098/0.009/0.061	0.143/0.219/0.021/0.146	0.170/0.244/0.023/0.177
0.8	0.127/0.203/0.019/0.129	0.078/0.118/0.011/0.082	0.108/0.143/0.013/0.126
1.2	0.211/0.328/0.030/0.220	0/0/0/0	0.035/0.036/0.003/0.051
1.6	0.322/0.481/0.044/0.347	0.112/0.151/0.014/0.131	0.079/0.123/0.012/0.085
1.9	0.472/0.616/0.058/0.547	0.251/0.304/0.031/0.346	0.225/0.286/0.028/0.281

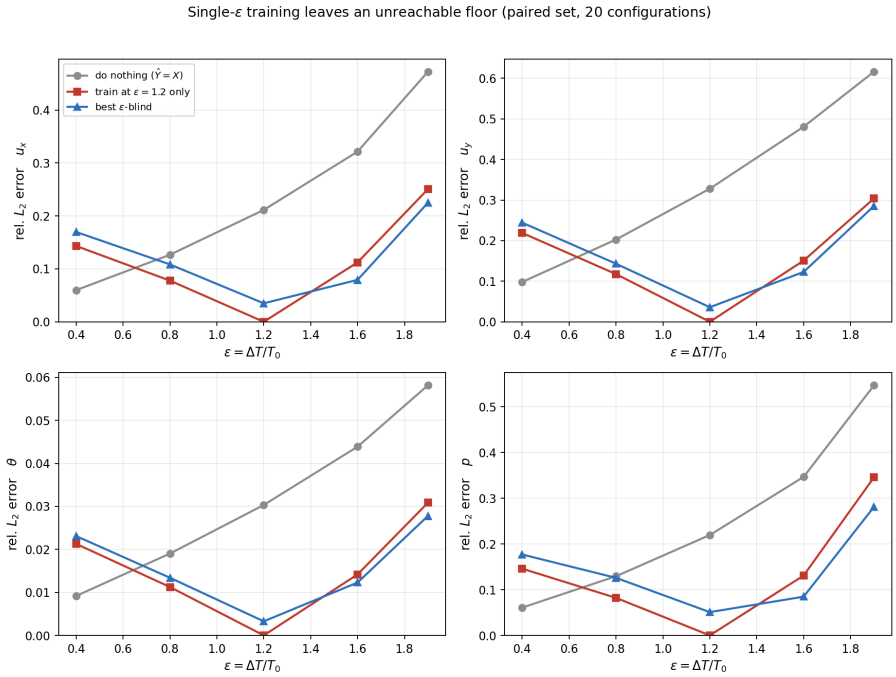


Figure 2: Single-ε training leaves an unreachable floor. Curves show "do nothing", training at $\varepsilon = 1.2$ only, and the best ε-blind field, for the 20 paired configurations.

5.4. Network baselines: the correction is information, not capacity

Table 6 reports a U-Net of width 32, 200 epochs, mean of three seeds, identical protocol for all columns (largest standard deviation over every cell: 0.0051).

Model A's error on the extrapolation split (0.2289) is within 2% of the ε-mean field of the paired set at $\varepsilon = 1.9$ (0.2249, Sec. 5.3), although the two are measured on different data sets; the agreement is consistent with a ceiling that is informational rather than architectural. Shallow learners plateau in the same place. A linear ridge regression and an RBF kernel ridge on the raw four-field input—the same information the network gets—reach u_x 0.2282 and 0.2222 in extrapolation, and adding ε as an extra column barely moves them (0.2188 and 0.2171, against 0.0937

Table 6: Surrogate ablation on the two fixed splits (U-Net, width 32, 200 epochs, mean of three seeds; largest standard deviation over all cells 0.0051). A sees only the four Boussinesq fields; B also receives ε as a constant input channel; P1 multiplies ε outside the network but still reads it as an input; P2 adds a learned quadratic factor; P3 removes ε from the input entirely. Errors are per-channel relative L_2 errors against Y. The curvature learned by P2 (seed 0) is $\kappa = 0.418/0.314/0.334/0.314$ (extrapolation) and $0.295/0.224/0.255/0.219$ (interpolation), for $u_x/u_y/\theta/p$; the three-seed means are $0.418/0.321/0.343/0.338$ and $0.286/0.232/0.252/0.265$. The own-data row of Table 9 is the P3 architecture trained with 150 epochs and one seed, the protocol of the public-transfer rows; this table uses 200 epochs and three seeds; the same architecture therefore differs between the tables by 2–6% in extrapolation and 6–12% in interpolation.

split	channel	A	B	P1	P2	P3
extrapolation ($n = 218$)	u_x	0.2289	0.1508	0.0937	0.0867	0.1214
	u_y	0.2733	0.1725	0.1060	0.0951	0.1277
	θ	0.0406	0.0255	0.0141	0.0131	0.0161
	p	0.2689	0.1791	0.1126	0.1007	0.1682
interpolation ($n = 200$)	u_x	0.1530	0.0673	0.0545	0.0507	0.0732
	u_y	0.1796	0.0714	0.0524	0.0488	0.0749
	θ	0.0261	0.0097	0.0078	0.0073	0.0092
	p	0.1792	0.0667	0.0454	0.0410	0.0893

for the structured model), because neither can represent a product of an ε-scale with a spatial field. A classical POD-ROM (44–52 modes, 99.9% of the snapshot energy) whose coefficients are regressed on $(\log_{10} Ra, \varepsilon)$ lands at 0.2937 with ε and 0.3001 without, i.e. at the level of doing nothing, and a purely linear surrogate that also sees the POD coefficients of X reaches only 0.2318. The plateau is not capacity—it is that the quantity to be added is absent from the input (`code/exp_shallow.py`, `code/exp_pod.py`).

A first-order perturbation baseline. If the correction is linear in ε, the obvious alternative is to solve $\mathcal{L}u_1 = S_1[X]$ once per query and scale. That costs one Jacobian assembly and one factorisation, a fraction of the 22.3 s nonlinear compressible solve; we quote the full solve (median 30 s here) as a conservative bound. We evaluate the accuracy of that response on the 20 configurations of the ε scan, using $P = (Y - X)/\varepsilon$ at $\varepsilon = 0.4$ as a finite-difference stand-in for the linear solution ($O(0.4)$ error in P), and predict $\hat{Y} = X + \varepsilon P$. The error is $5.3/5.5/0.5/5.7\%$ at $\varepsilon = 1.2$ and $22/26/2.9/26\%$ at $\varepsilon = 1.9$, against a do-nothing floor of $42/60/7.3/41\%$. The linear response removes at most half the gap and leaves the remainder to the $O(\varepsilon^2)$ term. Both approaches replace the same member, so the network's advantage is not speed: both save about the same end-to-end time. It is accuracy at $\varepsilon \to 1.9$, where the learned curvature κ keeps the part of the correction that the linear solve cannot produce (Table 6).

5.5. *The physics prior:* $D = \varepsilon P(X)$

Same backbone, only the output structure changes (Table 6 of Sec. 5.4; ε still read from a constant channel, target still D).

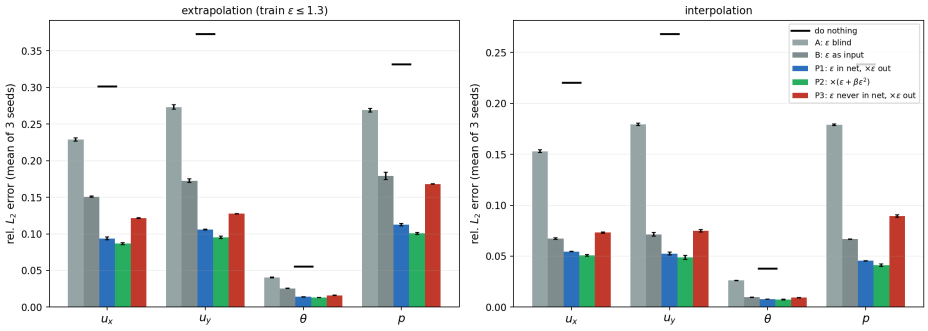


Figure 3: The prior ablation for extrapolation (left) and interpolation (right). A: ε-blind. B: ε as input. P1: ε in the network, multiplied outside. P2: learned $(\varepsilon + \kappa\varepsilon^2)$ factor. P3: ε never enters the network.

Three things happen at once (Table 6, Fig. 3). In extrapolation the factorised output beats a plain network that reads ε from a constant input channel by 1.61× (u_x) and 1.59–1.81× in the other channels, and the quadratic form adds a further 7.1–10.6% on top. That channel is a weak way to condition a network. A FiLM-conditioned U-Net, which modulates every block with ε, reaches 0.1001/0.1155/0.0164/0.1251 in extrapolation and 0.0544/0.0496/0.0073/0.0428 in interpolation, so against that baseline the factorised output gains 1.07× and the quadratic form 1.15× in u_x. Removing ε from the input as well costs accuracy relative to P1 and FiLM—0.1214 against 0.0937 and 0.1001 in u_x—but still beats the ε-blind network (1.89×); what it buys is the property P1 cannot have, a network that can be trained at one ε and applied at another (Sec. 5.9). Finally, the network *discovers* a positive curvature on its own, $\kappa = 0.22$–0.42, the same sign and order as the ε^2 term of the expansion in Sec. 3—an independent check of the asymptotic expansion.

Practical form. The prior costs two lines of code: train any regressor $\mathcal{N}$ on the ratio D/ε instead of on D, and at run time return $\hat{Y} = X + \varepsilon\,\mathcal{N}(X)$ after evaluating the Boussinesq member (or a surrogate for it) to obtain X. Supplying ε to $\mathcal{N}$ as well is the most accurate form (P1, 1.07× over a FiLM-conditioned network); leaving it out (P3) costs accuracy at fixed data but is what lets one network serve every contrast for which the prefactor is unchanged, because $\mathcal{N}$ then sees only X and re-learns the prefactor from the flow rather than from a tabulated value.

A Fourier neural operator (1.19 M parameters) shows the same pattern against its ε-blind version: u_x 0.2287 → 0.1334 with the prior and ε multiplied back outside the network (1.71×), statistically indistinguishable from an explicit ε input channel (0.1334 ± 0.0004 against 0.1342 ± 0.0008). On u_y and θ the prior is in fact better (0.1387 against 0.1456; 0.0177 against 0.0191); only on p does the ε channel win

(0.1536 against 0.1734). The prior's value is therefore not that it beats an ε-aware model—it is that it matches one while needing no ε in the input (Secs. 5.6, 5.9).

5.6. Data efficiency

The learning curve is tabulated in Table S4 of the supplementary material and plotted in Fig. 4.

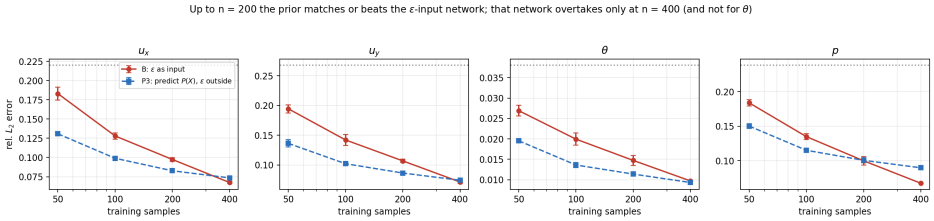


Figure 4: Learning curves. Up to $n = 200$ the prior matches or beats the ε-input network; the latter overtakes only at $n = 400$, and not for θ.

The naive expectation—that a structured model trades in-distribution accuracy for extrapolation—is wrong at small n. At $n = 50$ the prior is 1.4× better than the ε-input network (u_x 0.1308 against 0.1831, a 9σ separation across three seeds), and it is still ahead at $n = 200$ in u_x (7.3σ), u_y (7.6σ) and θ (3.6σ), with p already level (0.13σ) there. The crossover sits between $n = 200$ and 400 in u_x and u_y (log-linear interpolation: $n \approx 315$ and 360, i.e. near 300 in u_x), essentially at $n = 200$ in p, and θ never crosses—at $n = 400$ it is still 2.5σ behind. Concretely, the prior at $n = 200$ reaches what the ε-input network needs ≈ 270 samples (u_x) and 290 (u_y) to reach. So the prior is two things at once: the cheaper model when data is scarce, and—a model that needs no ε in its input, which is what makes a single-ε data set reusable.

5.7. External validation on an independent, published data set

The public release `NeuralConvection_2D_TrainTest` (v2; 4000 training pairs plus 200 test pairs, CC BY 4.0) was generated by another group, at a single $\varepsilon = 1.2$, $Ra \in [10^2, 3 \cdot 10^6]$ [5].

Conventions match. Recomputing their published `relL2_*` columns as $\|Y-X\|/\|X\|$ from the tensors reproduces them exactly (0.14233/0.13749/0.01356/0.19070 for sample 0); their pressure amplitude ratio is 0.929 against 0.960 for our fixed pairing, under the same median convention.

The law reproduces. On the $\varepsilon = 1.2$ slice:

Even the secondary trends reproduce. The correlation of R with $\log_{10} Ra$ is +0.133, −0.419, +0.647 and −0.198 on their $\varepsilon \equiv 1.2$ data and +0.032, −0.511, +0.662 and −0.263 on our $\varepsilon \in [1.1, 1.3]$ slice (64 samples, $u_x/u_y/\theta/p$). The θ channel—the one whose scatter we explain next—has $r = +0.66$ on ours and +0.65 on theirs, from independently generated data.

Table 7: External validation at $\varepsilon = 1.2$: public data against our own slice. Medians with p10–p90 in parentheses.

channel	public 4000 (median, p10–p90)	our data at $\varepsilon \in [1.1, 1.3]$ ($n = 64$)
u_x	0.1797 (0.112–0.300)	0.1839 (0.126–0.284)
u_y	0.2435 (0.141–0.403)	0.2250 (0.150–0.395)
θ	0.0228 (0.001–0.071)	0.0232 (0.001–0.056)
p	0.1858 (0.117–0.246)	0.1889 (0.128–0.262)

Five splits, four prefactors, one predictable pattern. The release publishes three further held-out splits generated by the same generator and the same solver commit (`v6_freefall_20260921`, `raw_both_deramp` pressure convention; the OOD split carries the later tag `v6_freefall_ood_20260925`): `test_in_distribution` (200 random-wall pairs), `test_ood` (200 random-wall pairs whose vertical-wall Fourier dials lie outside the training range) and `test_legacy` (200 pairs of the classical cavity: constant hot and cold side walls, adiabatic top and bottom). We had already downloaded `train_legacy`, 5000 classical-cavity pairs. Recomputing all of `relL2_u_x`, `relL2_u_y`, `relL2_T`, `relL2_p` for every sample of all five splits (38 400 values: 9600 samples × 4 channels) under our conventions reproduces the published numbers to within 1.2×10^{-7} (float32 round-off; `code/verify_conventions.py`), so nothing below is a convention artefact. At the fixed $\varepsilon = 1.2$ the medians of $R = \|Y - X\|/\|Y\|$ are listed in Table S5 of the supplementary material for all five splits, and in Table 7 for the two that matter here.

The exponent cannot be re-tested here—every split sits at one ε—but the prefactor is put under pressure and it moves: by 1.66× (u_x), 1.38× (u_y), 4.05× (θ) and 1.07× (p) between the classical cavity and the random-wall families at the same ε, and `train_legacy` and `test_legacy` agree to three digits in u_x and p (θ: 0.0926 vs 0.0836, 10%), which is the internal consistency check for the whole comparison. That is a statement about the flow, not a defect of the law, and the θ channel makes the mechanism explicit. Writing $\hat{u} = \max \|\boldsymbol{u}_X\|$ for the peak speed of the Boussinesq field, which is available to any surrogate, the θ-channel prefactor collapses on $\hat{u}$ (Fig. 5); the fits are in Table S6 of the supplementary material.

Two independent groups, two independent generators, six data sets: the random-wall families all give $c = 0.54$–0.69, the classical cavity—whose flow is a single strong circulation with $\hat{u}$ 41% above the public random-wall median and 56% above ours—gives $c = 2.06$–2.10 (`code/exp_uhat_law.py`). The same reading explains the residual scatter in $u_x/u_y/p$ in Sec. 5.1: A is a functional of the flow, and the flow is in the input.

5.8. What the correction does to the wall heat transfer

Every error quoted so far is a field error. For a heat-transfer application the quantity that matters is the Nusselt number, and the public classical cavity answers the question directly at $\varepsilon = 1.2$, because the two members share mesh and boundary

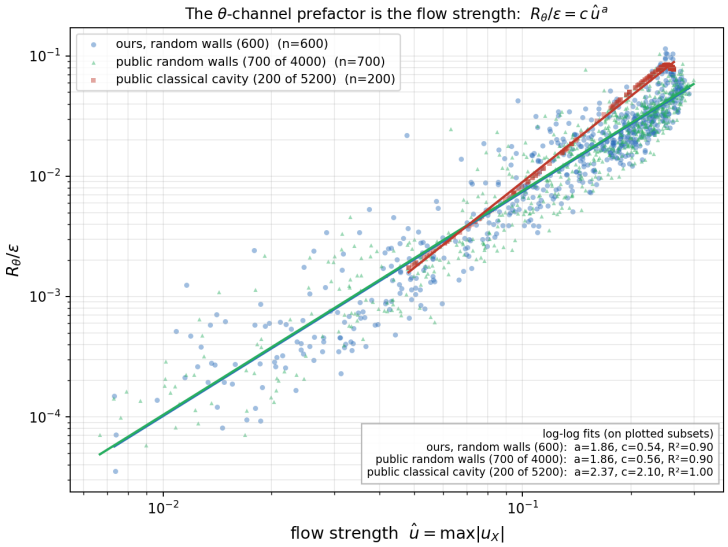


Figure 5: The θ-channel prefactor collapses on the peak Boussinesq speed $\hat{u}$ across two independent generators and two boundary-condition families.

conditions. We compute $Nu = -\langle \partial\theta/\partial n \rangle$ at the hot wall with a six-point one-sided stencil. On the Boussinesq member the same flux can be obtained without any wall stencil, from the horizontally integrated flux $\int [u_x\theta - (Ra\,Pr)^{-1/2}\partial_x\theta]\,dy$, which is independent of x to 0.1% at the three interior planes $x = 0.25, 0.5, 0.75$; the wall estimate agrees with it to 0.35%, which indicates the discretisation error of the estimator.

Table 8: Wall heat transfer at $\varepsilon = 1.2$ over the 200 classical-cavity samples of the public data set. The local change is $\Delta Nu(y)/Nu(y)$ along the hot wall. The integral change is the mean of the per-sample ratios at the six-point stencil, with the median and the 10–90% range; the ratio of the ensemble means is +0.67%. The stencil range is the two- to eight-point one-sided estimate of the wall gradient.

quantity	value	spread over samples
Nu, Boussinesq X	3.514	—
Nu, compressible Y	3.538	—
integral change $\Delta Nu/Nu$	+0.54%	median +0.58%, p10–p90 +0.06 to +0.92%; per-sample mean +0.35 to +0.62% over stencils
local change, L_1 norm over the wall	4.25%	p90 7.33%; unchanged for two to eight points
local change, pointwise	−0.65% (median)	p10–p90 −9.1 to +9.8%

The two statements are different in kind. The integral heat transfer is insensitive: the compressible member carries more heat in all 200 samples, by 0.01–0.94% (median 0.58%, 10–90% range 0.06–0.92%). The +0.54% of Table 8 is the mean of

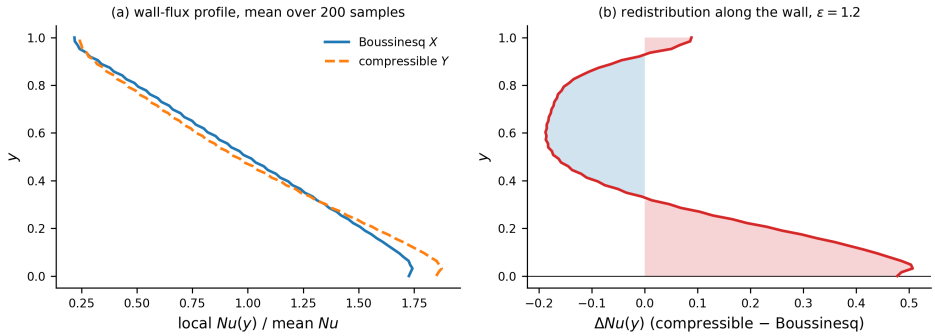


Figure 6: Wall heat transfer of the pair at $\varepsilon = 1.2$. (a) Local Nusselt number along the hot wall, normalised by its mean, for the two members. (b) Redistribution along the wall: the compressible member carries more heat in the lower half and less in the upper half, while the integral changes by less than 1%.

the per-sample ratios at the six-point stencil, and the ratio of the ensemble means is +0.67%. Across two- to eight-point stencils the per-sample mean spans +0.35 to +0.62% and the ratio of the ensemble means +0.30 to +0.90%. The local wall flux is not: the correction redistributes it by 4.25% in L_1 (p90 7.33%), with pointwise excursions of ±10%, and that number is the same for every stencil order. On average the correction adds heat over the lower half of the cavity and removes it over the upper half (Fig. 6b); the integral is the Nu-weighted mean of this redistribution, which is why it cancels. A surrogate trained on Boussinesq data therefore predicts the total heat load to better than 1% at this contrast and the local flux to only about 4%; which of the two matters is set by the application, not by the physics. The field statistic behind the cancellation is large: on the same 200 samples the temperature field itself differs by 8.4% in relative L_2, so an 8.4% change in the field produces a 0.6% change in its wall-normal derivative.

The gap grows with contrast. On the 20 configurations that carry five contrasts the mean of the per-sample relative gaps is +0.11% at $\varepsilon = 0.4$, +0.44% at 0.8, +1.09% at 1.2, +2.42% at 1.6 and +5.06% at 1.9, a log–log slope of 2.4; individual flows scatter around it (at $\varepsilon = 1.9$ the 10–90% range is −6.1 to +23%), because the sign of the local redistribution differs between flows. The heat load is therefore a second-order effect that reaches 5% only near $\varepsilon = 2$, but it is not zero, and a Boussinesq-trained surrogate cannot produce it: the three-orders-of-magnitude saving over the compressible solve is bought by keeping the compressible correction, not by replacing it with a cheaper Boussinesq run.

5.9. Cross-data-set transfer: the payoff, and its boundary

Two public single-ε data sets are available (Table 9, Fig. 7): 4000 random-wall pairs (our boundary-condition family) and 5000 classical-cavity pairs (a different one). Both sit at $\varepsilon \equiv 1.2$. We train the same U-Net on each—once with the plain output (target D), once with the prior (target D/ε, and ε multiplied back outside

the network)—and apply the result to our own 600 samples with no retraining, at ε from 0.2 to 1.9, values the training data never contained.

Table 9: Transfer from a single-ε public data set to our own $\varepsilon \in [0.2, 1.9]$ samples, on the extrapolation and interpolation test splits. Entries are $u_x/u_y/\theta/p$. The last row uses 382 training samples for the extrapolation column and 400 for the interpolation one. Both public sets are at $\varepsilon \equiv 1.2$. That row is the ε-blind prior (no ε input, 150 epochs, one seed), the same protocol as the public rows above; the P1/P3 columns of Table 6 use 200 epochs and three seeds.

trained on	model	extrapolation ($n = 218$)	interpolation ($n = 200$)
do nothing	$\hat{Y} = X$	0.3018/0.3731/0.0556/0.3320	0.2203/0.2683/0.0381/0.2386
public 4000, random walls	plain	0.1443/0.1730/0.0226/0.1838	0.1199/0.1441/0.0189/0.1435
public 4000, random walls	prior	0.0932/0.1085/0.0119/0.1322	0.0657/0.0769/0.0077/0.0891
our own training split	prior	0.1250/0.1321/0.0170/0.1719	0.0791/0.0811/0.0103/0.0947
public 5000, classical cavity	plain	0.2919/0.3711/0.0490/0.3192	0.2444/0.3612/0.0484/0.3217
public 5000, classical cavity	prior	0.3092/0.4730/0.0561/0.3280	0.2281/0.3498/0.0422/0.2584

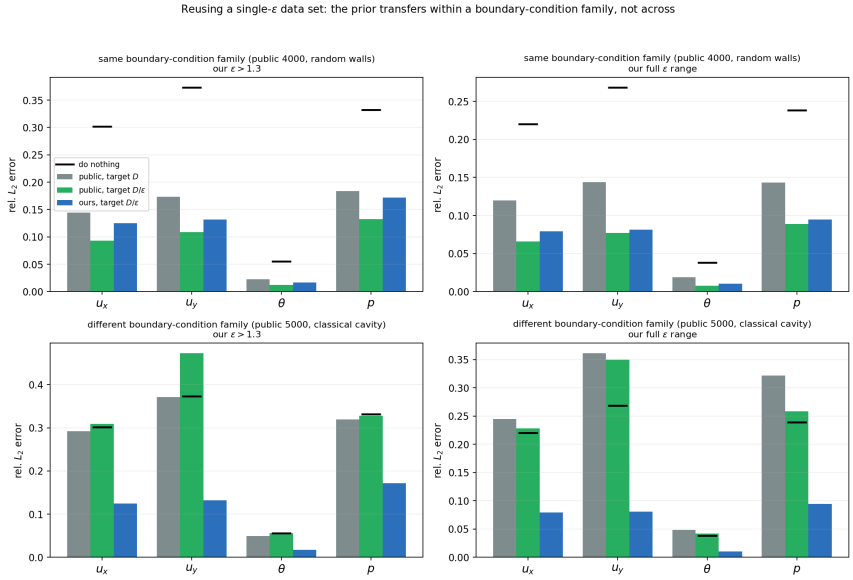


Figure 7: Reusing a single-ε data set. Within the boundary-condition family the prior transfers and beats the plain model; across families it fails, and can be worse than doing nothing.

Three readings, in order of how much they matter for practice.

- Within the family, the prior is what makes a single-ε data set reusable. The identical architecture, budget and data give 0.1443 (u_x) with the plain output and 0.0932 with the prior—a factor 1.55—and 3.2× better than doing nothing, on ε the data never contained. It is even better than the same prior trained on

our own 382 samples (0.1250), because it sees ten times more configurations. This is the concrete content of "the ε dependence need not be in the data".

- The prior also fixes the failure mode of the plain model. The plain model's error at unseen ε is largest where the training ε is furthest away (its correction is a fixed field, Sec. 5.3), whereas the prior's error stays proportional to how wrong P is, uniformly in ε.

- Across boundary-condition families the same trick fails, and the prior makes it worse. Trained on the classical cavity, the plain model lands at 0.2919 against a floor of 0.3018—it learned almost nothing transferable—and the prior model is *worse* than the plain one (0.3092 at $\varepsilon > 1.3$): it confidently multiplies a P learnt on the wrong flow by an ε that grows to 1.9. This is exactly what Sec. 5.7 predicts, and it is the same number: A is a functional of the flow, the classical cavity's flow has $\hat{u}$ 41% above the random-wall median, and a surrogate trained on that family predicts the wrong A. Reusing published data therefore requires the boundary-condition family to be represented; the law tells you in advance which reuse will work (`code/exp_public_transfer.py`, `code/exp_public_legacy_transfer.py`).

5.10. A second geometry (aspect ratio 2:1, 104 samples)

Same generator, same Ra and ε ranges, cavity twice as wide (Fig. 8). The ε exponent is unchanged in the ε-only fit ($m = 1.09/1.16/0.93/1.08$ for $u_x/u_y/\theta/p$, 0.7σ, 1.8σ, 0.3σ and 0.7σ away from the unit-square values; the two-parameter values are in Table S2), while the prefactor moves ($A = 0.175/0.175/0.008/0.222$ versus $0.147/0.193/0.010/0.148$). The Rayleigh trend is stronger for u_x and p here ($q = -0.14, -0.15$ vs $0.00, -0.03$) and reproduces for u_y and θ. The ε-scaling is geometry-independent; calibration of A is not—which is exactly why the surrogate takes X as input rather than a table of A. Changing the fluid instead (Pr 0.71 $\rightarrow$ 4.38) moves neither the exponent nor the prefactor by more than a few percent (Sec. 5.13).

5.11. Solver noise is not the source of the law

- Convergence quality over 600 samples (compressible member): median relative change 3.75×10^{-6}, p90 1.09×10^{-4}, worst 7.04×10^{-4}.

- Dropping the 65 samples with relative change $> 10^{-4}$ changes the u_x fit from $0.1809\varepsilon - 0.0212$ to $0.1813\varepsilon - 0.0221$—unchanged.

- Re-running the 10 worst samples with `steady_tol` tightened from 10^{-6} to 10^{-8} leaves the solution bit-identical ($|\Delta Y|/|Y| = 0$). The reason is structural: the compressible march terminates on the zero-iteration guard, not on `steady_tol`, so the residual reported is the fixed-point residual of the discrete solution, not unfinished iteration.

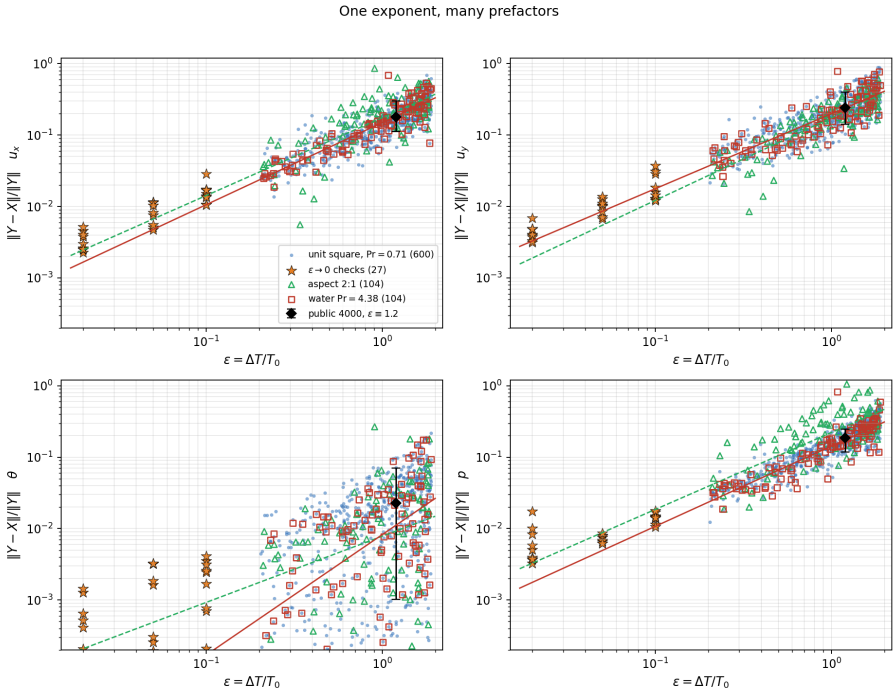


Figure 8: One exponent, many prefactors: unit square ($Pr = 0.71$), aspect 2:1, $Pr = 4.38$, and the public data set at $\varepsilon = 1.2$.

Table 10: Wall-clock cost, 64^2 mesh. Solver rows use eight CPU threads; the surrogate runs on an RTX 4060 Laptop GPU (batch 1) or one CPU thread. The two members are timed separately over the 600 samples; the last two rows are totals that combine the measured medians.

	median	mean	p90	max
paired solve, 600 samples	33.9 s	38.0 s	59.9 s	510 s
Boussinesq member alone	11.7 s	13.1 s	21.7 s	—
compressible member alone	22.3 s	25.0 s	37.3 s	—
paired solve, 100-sample ε scan	55.8 s	67.7 s	136 s	202 s
surrogate, GPU, batch 1	6.3 ms			
surrogate, GPU, batch 256	0.49 ms/sample			
surrogate, CPU, 1 thread	14.0 ms			
end-to-end pair (Boussinesq + surrogate)	11.7 s			
100-contrast sweep, total	12.3 s			

5.12. Cost and speed-up

Table 10 gives the wall-clock cost. A paired solve splits into a Boussinesq member (median 11.7 s) and a compressible member (22.3 s); the surrogate replaces only the second. Per pair the cost therefore falls from 33.9 s to 11.7 s, a factor 2.9, and the comparison of one inference with one compressible solve (22.3 s/6.3 ms = 3.5×10^3) describes only that member. The regime that matters for a parameter study is the amortised one: X does not depend on ε, so N contrasts cost one Boussinesq solve plus N inferences. For $N = 100$ that is 12.3 s against 3.39×10^3 s, a factor 2.8×10^2, and the Boussinesq solve is then the limit. The solver runs on eight CPU threads and the surrogate on a GPU; the two are not the same hardware.

5.13. A third Prandtl number: $Pr = 4.38$

Pr and ε do not mix. Pr enters the coefficients of both members ($Re^{-1} = (Pr/Ra)^{1/2}$) and hence the Boussinesq solution X, but the ε-expansion of Sec. 3 never multiplies the two. The prediction is therefore sharp and asymmetric: the exponent must not move, while the prefactor may, because X itself has changed.

We generated 104 further pairs at $Pr = 4.38$, the Prandtl number of water near 40°C, keeping the equation of state, γ and the transport coefficients of the air run: the study isolates the Pr axis and does not claim water properties (Table 11, Fig. 9). The pairs are bit-identical to the first 104 samples of the main batch in every other respect: same seed stream, hence the same wall-temperature Fourier draws, the same Ra and the same ε. The two families are therefore a strict paired experiment in which the fluid is the only difference (the Mach-number law is unchanged, and Ma is inert, Sec. 2.2).

Table 11: The fluid axis: 104 strictly paired configurations at $Pr = 0.71$ and $Pr = 4.38$ (same seed stream, same Ra and ε). The exponent is unchanged; the paired median ratio of R is 0.95–1.01, while the fitted prefactor of θ moves by 15% because the fit re-weights the ensemble.

channel	A, $Pr = 0.71$	A, $Pr = 4.38$	m, $Pr = 0.71$	m, $Pr = 4.38$
u_x	0.1468	0.1496	1.157	1.150
u_y	0.1934	0.1985	1.047	1.049
θ	0.0097	0.0082	1.668	1.686
p	0.1482	0.1426	1.109	1.124

On the 104 matched configurations the paired ratios $R(Pr = 4.38)/R(Pr = 0.71)$ have medians 1.006, 0.999, 0.999 and 0.951 ($u_x/u_y/\theta/p$; p10–p90 0.99–1.12, 0.90–1.03, 0.95–1.06, 0.82–1.00), and the paired exponents agree to 1–2%: 1.150 vs 1.157, 1.049 vs 1.047, 1.686 vs 1.668, 1.124 vs 1.109. A 6.2-fold change in Prandtl number leaves the discrepancy of the same configuration unchanged to 0.1% in θ, 0.1% in u_x and u_y, and 5% in p.

Pr is therefore not a break-symmetry axis either, and the law's two ingredients separate cleanly: ε sets the exponent, the flow sets the prefactor, and the fluid

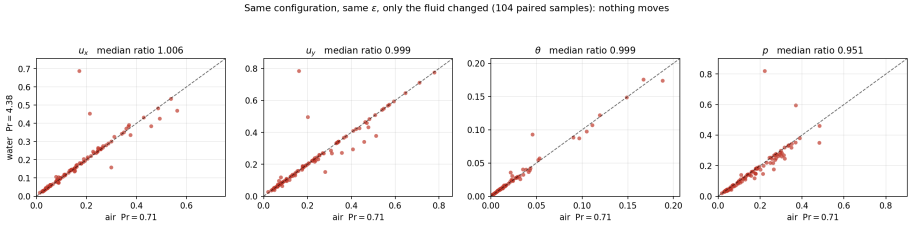


Figure 9: Same configuration, same ε, only the Prandtl number changed: $Pr = 0.71$ against $Pr = 4.38$, 104 paired samples. The panel labels follow the original run names; both runs use the same equation of state and transport coefficients, and only Pr differs.

changes neither by more than a few percent on the paired comparison. Keeping the strict pairing matters for the last digit—the unpaired fits give θ 0.0097 $\rightarrow$ 0.0082 (a 15% move, the same direction as the 5% paired move) and u_x/u_y +2–3%—which is the same statement as Sec. 5.10 (a geometry moves A) and Sec. 5.7 (boundary-condition families move it by up to 4×).

5.14. A three-dimensional check

The law is calibrated in two dimensions. To test whether the exponent survives in three, we generated twelve paired steady solutions with the same generator: three fixed flows—a three-cell Voronoi wall pair, a two-cell wall pair and a constant-wall cavity—each solved at $\varepsilon = 0.2$, 0.4, 0.8 and 1.2, with $Ra = 10^4$, $Pr = 0.71$ and $res = 16$ (33^3 velocity nodes, 17^3 temperature and pressure nodes). The pairing is exact: within each flow the wall temperatures are the same seed stream scaled affinely about T_0, so the low-fidelity member is the same solution up to the ε-blind rescaling of Sec. 2.3.

Table 12 and Fig. 10 give the result. In every channel and every flow a power law fits the four ε values in log–log coordinates with $R^2 > 0.998$, and the per-flow exponents are 1.063 ± 0.019 (u_x), 1.033 ± 0.012 (u_y), 1.092 ± 0.015 (θ) and 1.100 ± 0.010 (p)—within 0.11 of unity and the flow-mean exponents within 0.10 of the two-dimensional full-window fit of Table 4. The comparison is to the full-window fit rather than to a 2-D fit restricted to the same window: the 2-D samples are drawn at random ε and their prefactor varies by up to 4× between flows (Sec. 5.7), so a narrow-window regression there measures that scatter rather than the local slope. The prefactor is not universal here either: across the three flows it varies by 3.8× (u_x), 2.0× (u_y), 10× (θ) and 2.0× (p), which is the same statement as Sec. 5.7, that A is a functional of the flow. What transfers to three dimensions is the exponent, which is what the factorisation predicts.

A POD-ROM check of the prior. The same twelve samples also support a linear surrogate check. We train a four-mode POD-ROM on three flows and three contrasts (nine samples) and test on the held-out contrast (three samples), repeating for each of the four contrasts. The plain ROM predicts D from $[1, \varepsilon, \boldsymbol{c}_X]$, where $\boldsymbol{c}_X$ are four

POD coefficients of the Boussinesq field; the prior ROM predicts $P = D/\varepsilon$ from $[1, \boldsymbol{c}_X]$ and multiplies by ε outside. Four modes are fewer than the nine training samples. Averaged over the four held-out contrasts, the prior lowers the relative error in every channel (Table 13): u_x from 0.118 to 0.054, u_y from 0.136 to 0.110, θ from 0.024 to 0.019 and p from 0.108 to 0.064. The gain is largest at $\varepsilon = 0.2$ and 0.4 (3–10× in u_x) and is not uniform: at $\varepsilon = 0.8$ the ε-aware plain ROM is better in all four channels. We read this as a check that the factorisation remains useful in three dimensions at small contrast, not as a claim of uniform superiority.

The check is limited to $Ra = 10^4$, $res = 16$ and $\varepsilon \leq 1.2$. The twelve pairs cost 13–29 min each (median 16 min) of outer wall-clock time on this machine. We therefore report the exponent transfer, not a three-dimensional sweep.

Table 12: Three-dimensional check: twelve strictly paired samples, three fixed flows × four ε at $Ra = 10^4$, $Pr = 0.71$, $res = 16$. Per-flow exponents are power-law fits to the four ε values of that flow. Flows are ordered three-cell Voronoi, two-cell Voronoi, constant wall. The prefactor is listed per flow to show that it does not transfer between flows.

channel	m (mean±s.d. over flows)	per-flow m	A per flow
u_x	1.063 ± 0.019	1.09/1.06/1.05	0.066/0.180/0.251
u_y	1.033 ± 0.012	1.04/1.02/1.04	0.121/0.158/0.247
θ	1.092 ± 0.015	1.10/1.10/1.08	0.006/0.023/0.063
p	1.100 ± 0.010	1.10/1.11/1.09	0.094/0.177/0.183

Table 13: Three-dimensional POD-ROM check on the twelve paired samples. Leave-one-contrast-out: train on three flows × three ε (nine samples), test on the held-out ε (three samples), repeated for the four contrasts. Entries are mean relative L_2 errors over the four folds; the ROM uses four POD modes and ridge regression ($\lambda = 10^{-4}$). The gain is the plain-ROM error divided by the prior-ROM error.

channel	do nothing	plain ROM	prior ROM	gain
u_x	0.1406	0.1181	0.0542	2.18
u_y	0.1376	0.1357	0.1100	1.23
θ	0.0261	0.0244	0.0193	1.26
p	0.1136	0.1078	0.0644	1.67

6. Limitations

1. Two dimensions for the main sweep. The law is calibrated on 2-D solutions. The twelve-sample 3-D check of Sec. 5.14 reproduces the exponent, but covers one Rayleigh number, $res = 16$ and $\varepsilon \leq 1.2$; the POD-ROM baseline is a four-mode linear model, not a deep operator. Each 3-D pair costs 13–29 min of outer wall-clock time (median 16) on this machine. A 3-D sweep at higher Ra or resolution remains out of reach here.

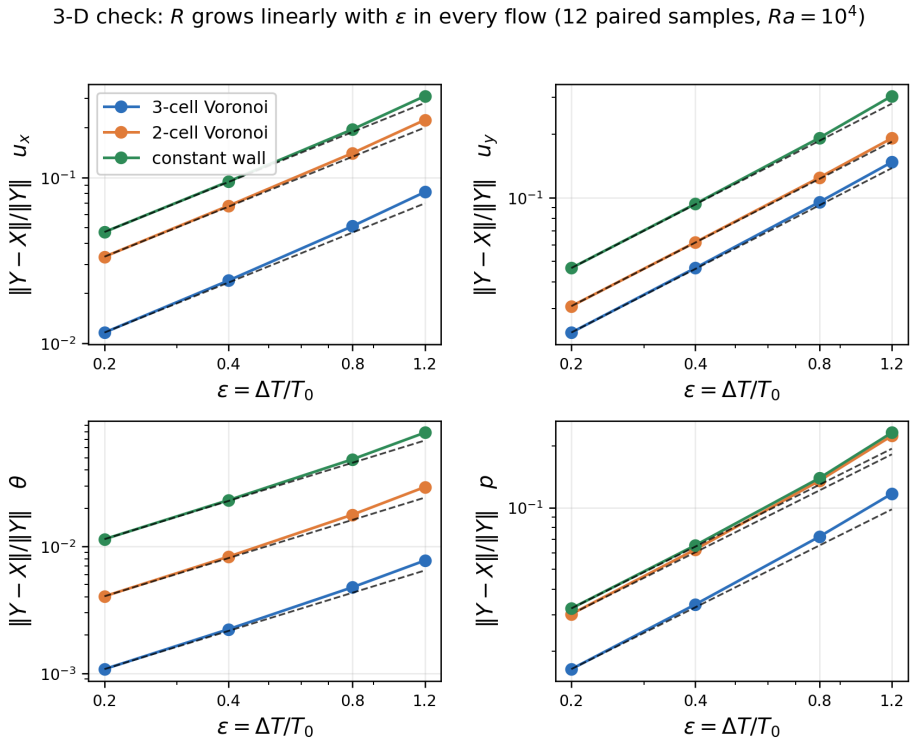


Figure 10: The three-dimensional check in log–log coordinates: twelve paired samples, three fixed flows and four temperature contrasts. Dashed lines are a slope-one reference anchored at the lowest ε of each flow.

2. *Ma* is inert here, so the break-symmetry group is one-dimensional rather than two. At larger *Ma* this would have to be redone.
3. $\varepsilon < 2$, imposed by the equation of state (the cold wall must stay positive).
4. The two-parameter fits of Sec. 5.1 leave 26/23/28% of the variance unexplained in $u_x/u_y/\theta$ and 17% in p; the $\hat{u}$ form of Sec. 5.7 closes that gap for θ and leaves the same residual flow-dependence in the other three channels.
5. The prefactor A is configuration-dependent and must be learned: it moves with the geometry (Sec. 5.10) and by 1.66× (u_x) / 1.38× (u_y) / 4.05× (θ) / 1.07× (p) between boundary-condition families of an independent data set (Sec. 5.7), while it is insensitive to the fluid (*Pr*, Sec. 5.13).
6. The public data set and ours differ by 0.08% in Ra_{eff} at $\varepsilon = 1.2$ ($1.2/1.095^2 = 1.000813$); their v2 release corrects this and the two now agree.

7. Conclusion

We set out to build a surrogate for compressible natural convection and found that the object it has to reproduce—the gap between the Boussinesq solution and the

compressible one—is governed by a single parameter in an exact and predictable way.

1. One parameter. In this non-dimensionalisation $Fr^2 = \Delta T/T_0$ identically, so the break-symmetry axis is $\varepsilon = \Delta T/T_0$ and nothing else; Mach-number effects are $O(Ma^2)$ and are invisible at the values reached here.
2. A clean law. $R = \|Y - X\|/\|Y\| = A\varepsilon^m$ with $m = 1.023 \pm 0.018$, 0.995 ± 0.019, 1.024 ± 0.077, 1.009 ± 0.015 (u_x, u_y, θ, p), an intercept indistinguishable from zero, the exponent quoted over the full two-decade window (from $\varepsilon = 0.8$ upwards the same points give 1.23–1.32, the ε^2 curvature of Sec. 5.1), and a 134-family boundary-condition description (133 dummies) that adds only 2–3 pp of adjusted R^2 to the velocity and pressure channels. The exponent is geometry-independent and fluid-independent; only the prefactor A must be calibrated, which is why the surrogate takes X as input rather than a table of A. Twelve paired 3-D solutions reproduce the exponent (flow means 1.03–1.10, per-flow 1.02–1.11; Sec. 5.14).
3. A reason. The ε-expansion of the compressible member reproduces the Boussinesq system at $O(1)$ and produces the correction through $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot (\theta_0 \boldsymbol{u}_0)$ and the density-modulated inertia. The linearised operator does not contain ε, so the correction factorises, $D = \varepsilon P(X) + \varepsilon^2 Q(X) + \ldots$, with a shape that does not rotate. All five consequences of the expansion are confirmed numerically.
4. A design rule. Predicting $P(X)$ and multiplying by ε outside the network—two lines of code—converts the law into a prior that (i) beats a FiLM-conditioned network by 1.07–1.15×, (ii) lets the network discover the ε^2 term itself, (iii) works the same way for a Fourier neural operator, and (iv) makes a public data set that exists at a single temperature contrast reusable at others—0.093 (u_x) on $\varepsilon > 1.3$ against 0.144 for the plain model and 0.302 for doing nothing—which is the difference between a surrogate that must be retrained and one that can be applied. The scope of that reuse is set by Sec. 5.7: it holds within a boundary-condition family and fails across one, predictably.

The same factorisation lowers the mean u_x error of a four-mode POD-ROM on the 3-D samples from 0.118 to 0.054 over four held-out contrasts.

Section 6 lists the limits: two dimensions for the main sweep, $\varepsilon < 2$, one-dimensional break symmetry, and a prefactor whose two-parameter fits leave 26% (u_x), 23% (u_y) and 28% (θ) of the variance unexplained. None of them touch the claim that matters for practice, which is that the ε dependence is known in closed form and should not be left to the network to discover from data.

8. Reproducibility

Table S7 of the supplementary material maps each script to the results it produces. All scripts are in the `code/` directory of the project (executed in WSL2 with

dolfinx 0.7.3 for the solver and PyTorch 2.10 for the surrogates); raw data, logs and JSON results are under `~/nc_run/`. The 3-D POD-ROM check is `code/exp_3d_rom_eps.py`, with machine-readable output `èőžæŰĞ/3d_rom_eps.json`.

Code and every machine-readable result are mirrored at `https://github.com/sheep-cloud12138/Algorithms-instead-of-simulations` (working branch `dev`; `master` carries snapshots that have passed an independent review). The raw solver output (600 + 104 + 104 + 27 two-dimensional pairs and 12 three-dimensional pairs, ∼6 GB) stays on the machine described above; every number in this paper is regenerated from the JSON results in `~/nc_run/p7/` by reporting scripts, not transcribed by hand.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used an AI assistant (OpenAI Codex) to develop analysis scripts, to draft and edit the manuscript, and to check numerical consistency between the text and the stored results. After using this tool, the authors reviewed and edited the content and take full responsibility for the content of the publication.

Data availability

The public paired data set of Menglik et al. [5] is available under CC BY 4.0. Our generated data, solver configuration, and every script that reproduces the numbers in this paper are available at `https://github.com/sheep-cloud12138/Algorithms-instead-of-simulations` (branch `dev`); the raw solver output is available from the authors on request.

References

[1] M. Giacomini, P. Díez, Surrogates for physics-based and data-driven modelling of parametric systems: Review and new perspectives, Archives of Computational Methods in Engineering 33 (2026) 8519–8551. doi:10.1007/s11831-026-10552-4.

[2] C. Meng, S. Griesemer, D. Cao, S. Seo, Y. Liu, When physics meets machine learning: A survey of physics-informed machine learning, Machine Learning for Computational Science and Engineering 1 (2025) 20. doi:10.1007/s44379-025-00016-0.

[3] E. A. Spiegel, G. Veronis, On the Boussinesq approximation for a compressible fluid, The Astrophysical Journal 131 (1960) 442–447. doi:10.1086/146849.

[4] D. D. Gray, A. Giorgini, The validity of the Boussinesq approximation for liquids and gases, International Journal of Heat and Mass Transfer 19 (1976) 545–551. doi:10.1016/0017-9310(76)90168-X.

[5] N. Menglik, A. Shao, D. Hyde, A neural surrogate approach for simulating natural convection problems, 2026. arXiv:2606.25259.

[6] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar, Fourier neural operator for parametric partial differential equations, in: International Conference on Learning Representations, 2021. ArXiv:2010.08895.

[7] Q. Huhn, M. Tano, J. C. Ragusa, Y. Choi, Parametric dynamic mode decomposition for reduced order modeling, Journal of Computational Physics 475 (2022) 111852. doi:10.1016/j.jcp.2022.111852.

[8] F. Feng, Y. Li, Z. Chen, W.-T. Wu, J.-Z. Peng, M. Mei, Graph convolution network-based surrogate model for natural convection in annuli, Case Studies in Thermal Engineering 57 (2024) 104330. doi:10.1016/j.csite.2024.104330.

[9] J.-Z. Peng, N. Aubry, Y. Li, M. Mei, Z. Chen, W.-T. Wu, Physics-informed graph convolutional neural network for modeling geometry-adaptive steady-state natural convection, International Journal of Heat and Mass Transfer 216 (2023) 124593. doi:10.1016/j.ijheatmasstransfer.2023.124593.

[10] L. Lu, P. Jin, G. Pang, Z. Zhang, G. E. Karniadakis, Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators, Nature Machine Intelligence 3 (2021) 218–229. doi:10.1038/s42256-021-00302-5.

[11] S. Wang, H. Wang, P. Perdikaris, Learning the solution operator of parametric partial differential equations with physics-informed DeepONets, Science Advances 7 (2021) eabi8605. doi:10.1126/sciadv.abi8605.

[12] X. Zhao, X. Chen, Z. Gong, W. Zhou, W. Yao, Y. Zhang, RecFNO: A resolution-invariant flow and heat field reconstruction method from sparse observations via Fourier neural operator, International Journal of Thermal Sciences 195 (2023) 108619. doi:10.1016/j.ijthermalsci.2023.108619.

[13] X. F. Yu, W. Gao, K. Guo, Z. Zhang, H. Wang, M. Wang, S. Qiu, W. Tian, Fourier neural operator-based approach for real-time prediction of high-temperature heat pipes frozen startup, Applied Thermal Engineering 302 (2026) 131964. doi:10.1016/j.applthermaleng.2026.131964.

[14] I. Şahin, C. Moya, A. Mollaali, G. Lin, G. Paniagua, Deep operator learning-based surrogate models with uncertainty quantification for optimizing internal cooling channel rib profiles, International Journal of Heat and Mass Transfer 219 (2023) 124813. doi:10.1016/j.ijheatmasstransfer.2023.124813.

[15] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics 378 (2019) 686–707. doi:10.1016/j.jcp.2018.10.045.

[16] S. Cai, Z. Wang, S. Wang, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks for heat transfer problems, ASME Journal of Heat and Mass Transfer 143 (2021) 060801. doi:10.1115/1.4050542.

[17] D. Jalili, S. Jang, M. Jadidi, G. Giustini, A. Keshmiri, Y. Mahmoudi, Physics-informed neural networks for heat transfer prediction in two-phase flows, International Journal of Heat and Mass Transfer 221 (2023) 125089. doi:10.1016/j.ijheatmasstransfer.2023.125089.

[18] T. De Ryck, S. Mishra, Numerical analysis of physics-informed neural networks and related models in physics-informed machine learning, Acta Numerica 33 (2024) 633–713. doi:10.1017/S0962492923000089.

[19] K. Luo, J. Zhao, Y.-P. Wang, J. Li, J. Wen, J. Liang, H. Soekmadji, S. Liao, Physics-informed neural networks for PDE problems: A comprehensive review, Artificial Intelligence Review 58 (2025) 323. doi:10.1007/s10462-025-11322-7.

[20] P. K. Jha, Residual-based error corrector operator to enhance accuracy and reliability of neural operator surrogates of nonlinear variational boundary-value problems, Computer Methods in Applied Mechanics and Engineering 419 (2023) 116595. doi:10.1016/j.cma.2023.116595.

[21] L. Cao, T. O'Leary-Roseberry, P. K. Jha, J. T. Oden, O. N. Ghattas, Residual-based error correction for neural operator accelerated infinite-dimensional Bayesian inverse problems, Journal of Computational Physics 486 (2023) 112104. doi:10.1016/j.jcp.2023.112104.

[22] M. Zhu, H. Zhang, A. Jiao, G. E. Karniadakis, L. Lu, Reliable extrapolation of deep neural operators informed by physics or sparse observations, Computer Methods in Applied Mechanics and Engineering 412 (2023) 116064. doi:10.1016/j.cma.2023.116064.

[23] S. Goswami, K. Kontolati, M. D. Shields, G. E. Karniadakis, Deep transfer operator learning for partial differential equations under conditional shift, Nature Machine Intelligence 4 (2022) 1155–1164. doi:10.1038/s42256-022-00569-2.

[24] P. Conti, M. Guo, A. Manzoni, J. S. Hesthaven, Multi-fidelity surrogate modeling using long short-term memory networks, Computer Methods in Applied Mechanics and Engineering 404 (2022) 115811. doi:10.1016/j.cma.2022.115811.

[25] W. Chen, P. Stinis, Feature-adjacent multi-fidelity physics-informed machine learning for partial differential equations, Journal of Computational Physics 498 (2023) 112683. doi:10.1016/j.jcp.2023.112683.

[26] Y. Lyu, X. Zhao, Z. Gong, X. J. Kang, W. Yao, Multi-fidelity prediction of fluid flow based on transfer learning using Fourier neural operator, Physics of Fluids 35 (2023) 077118. doi:10.1063/5.0155555.

[27] G. Ahlers, S. Großmann, D. Lohse, Heat transfer and large scale dynamics in turbulent Rayleigh–Bénard convection, Reviews of Modern Physics 81 (2009) 503–537. doi:10.1103/RevModPhys.81.503.

[28] H. Yik, V. Valori, S. Weiss, Turbulent Rayleigh–Bénard convection under strong non-Oberbeck–Boussinesq conditions, Physical Review Fluids 5 (2020) 103502. doi:10.1103/PhysRevFluids.5.103502.

[29] M. Macek, G. Zinchenko, V. Musilová, P. Urban, J. Schumacher, Assessing non-Oberbeck–Boussinesq effects of convection in cryogenic helium, Physical Review Fluids 8 (2023) 094606. doi:10.1103/PhysRevFluids.8.094606.

[30] X. Pan, J.-I. Choi, Non-Oberbeck–Boussinesq effects on a water-filled differentially heated vertical cavity, Physics of Fluids 35 (2023) 113609. doi:10.1063/5.0173049.

[31] S.-N. Xia, Z.-H. Wan, S. Liu, Q. Wang, D.-J. Sun, Flow reversals in Rayleigh–Bénard convection with non-Oberbeck–Boussinesq effects, Journal of Fluid Mechanics 798 (2016) 628–642. doi:10.1017/jfm.2016.338.

[32] J. Zhang, Y. Wu, N. Gui, X. Yang, J. Tu, S. Jiang, Investigation of compressible flow under low Mach number in an enclosed square cavity with a novel non-Boussinesq algorithm, Physics of Fluids 35 (2023) 123120. doi:10.1063/5.0175738.

[33] H. Satake, T. Tagawa, Mathematical modeling for low-Mach-number compressible thermal convection analysis in a square enclosure, Physics of Fluids 36 (2024) 127168. doi:10.1063/5.0243786.

[34] P. Bella, E. Feireisl, F. Oschmann, Rigorous derivation of the Oberbeck–Boussinesq approximation revealing unexpected term, Communications in Mathematical Physics 403 (2023) 1245–1273. doi:10.1007/s00220-023-04823-5.

[35] Y. Song, Y. Wu, Y. Wang, X. Yang, DimINO: Dimension-informed neural operator learning, 2024. arXiv:2410.05894.